



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS

2011 Year 8 Assessment Task 3

General instructions

- Working time – 50 minutes.
- Marks may be deducted for careless or poorly arranged work.
- Commence each new question on a new sheet.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 8M1 – Ms Everingham/Mr Berry
- ☐ 8M2 – Mr Weiss/Mr Berry
- ☐ 8M3 – Mr Lin/Mrs Collins
- ☐ 8M4 – Mrs Ziaziaris
- ☐ 8M5 – Mr Ireland/Mrs Collins

NAME: # SHEETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	Total	%
MARKS	$\overline{14}$	$\overline{15}$	$\overline{10}$	$\overline{11}$	$\overline{10}$	$\overline{60}$	

Question 1 (14 Marks)	Commence a NEW page.	Marks
(a) Expand and simplify:		
i. $(5y - 2)(2y + 3)$.		2
ii. $(3b + 1)^2$		2
(b) Factorise fully:		
i. $3y^2 - 2y$.		1
ii. $x(a + b) + y(a + b)$.		1
iii. $d^2 - 81$.		1
iv. $a^2 + 2a - 15$.		1
v. $x^4 - 1$.		2
(c) Simplify fully:		
i. $\frac{3x}{2} + \frac{5x}{7}$.		1
ii. $\frac{3x + 21}{3}$.		1
iii. $\frac{5a + 4b}{10a + 8b}$.		2

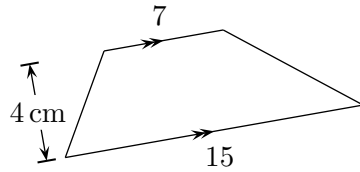
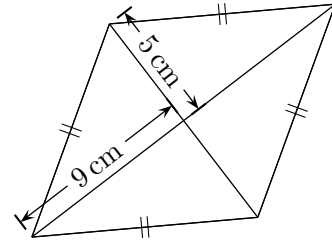
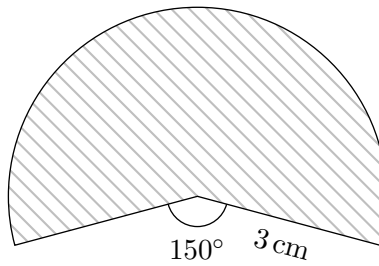
Question 2 (15 Marks)

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Marks

(a) Find the area of each of the following shapes:

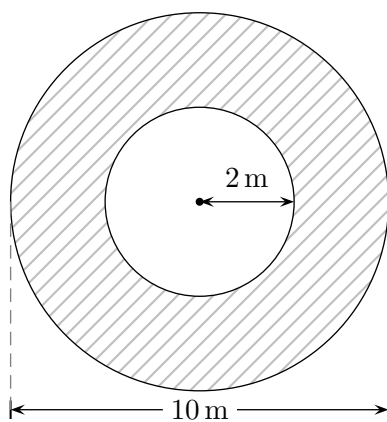
i.

**2** ii.**2**(b) The area of a circle is 3.0512 m^2 . Find the radius and circumference. Give both your answers correct to 2 decimal places.**3**(c) Find the *exact* area of the shaded sector.**3****Question 2 continued on the next page ...**

Question 2 continued from the previous page ...

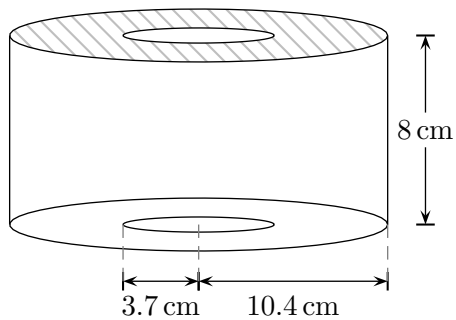
- (d) Find the shaded area.

2



- (e) Find the volume, correct to 2 decimal places.

3



Question 3 (10 Marks)

Commence a NEW page.

Marks

- (a) Refer to Table 1 for the following question.

Country	Area (km ²)	Population	Members in European Parliament
Austria	88 850	7 796 000	21
Belgium	30 510	10 131 000	25
Denmark	43 070	5 216 000	16
Finland	338 130	5 099 000	16
France	551 500	57 800 000	87
Germany	356 910	81 338 000	99
Greece	131 990	10 400 000	25
Ireland	70 280	3 560 000	15
Italy	301 270	56 930 000	87
Luxembourg	2 590	407 000	6
Netherlands	41 526	15 424 000	31
Portugal	92 390	9 902 000	25
Spain	504 780	39 190 000	64
Sweden	449 960	8 816 000	22
United Kingdom	243 368	58 784 000	87

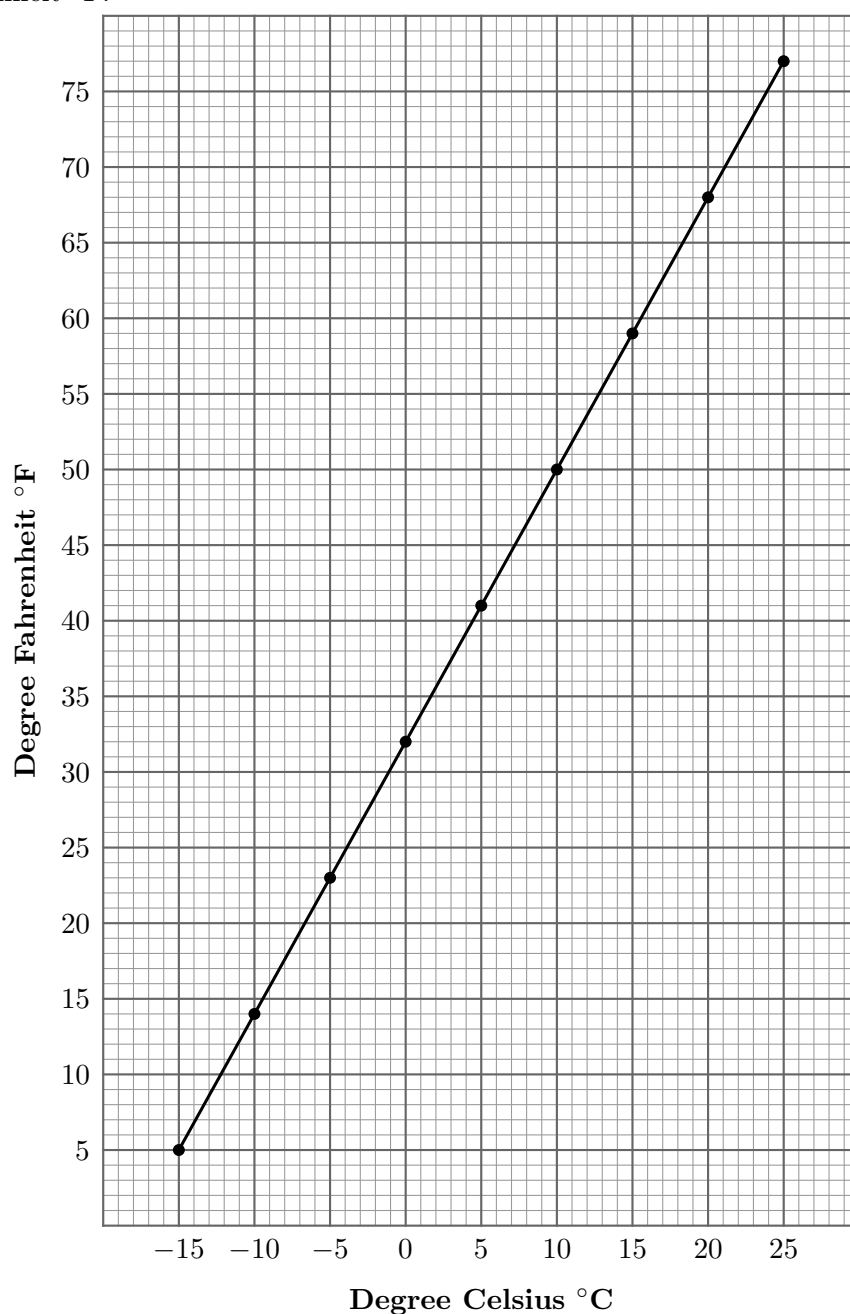
Table 1: Countries in the European Union.

- i. Which European Union country has the largest population? **1**
 - ii. To the nearest whole number, how many times larger is the largest European Union country than the smallest? **1**
 - iii. To the nearest whole number, what is the population density of Spain, in people/km²? **1**
- (b) In a survey, people were asked for their favourite brand of coffee. 32 preferred Nescafe, 80 Moccona, 64 Bushell and 16 Maxwell House. Represent this information on a sector graph with a radius of 6 cm. **3**

Question 3 continued on the next page ...

Question 3 continued from the previous page ...

- (c) The line graph below may be used to convert degrees Celsius $^{\circ}\text{C}$ to degrees Fahrenheit $^{\circ}\text{F}$.



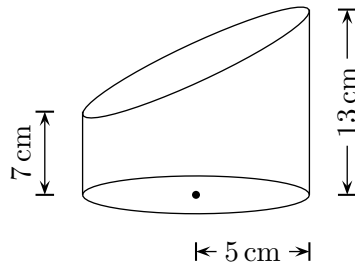
- | | |
|---|---|
| i. Use the graph to convert 10°C to Fahrenheit $^{\circ}\text{F}$. | 1 |
| ii. Convert -5°C to $^{\circ}\text{F}$. | 1 |
| iii. Convert 32°F to $^{\circ}\text{C}$. | 1 |
| iv. Kate's Facebook friend in Toronto Canada, posted on Kate's profile and said the temperature was 0°F . Use the graph to find the approximate temperature in $^{\circ}\text{C}$. | 1 |

Question 4 (11 Marks)

Commence a NEW page.

Marks

- (a) Find the volume of the solid.

3

- (b) Find the volume of a cylinder of radius 8 cm and perpendicular height 10 cm.

2

- (c) A right prism has a square base and a height of 12 cm. If its volume is
- 300 cm^3
- , find the surface area of the prism.

3

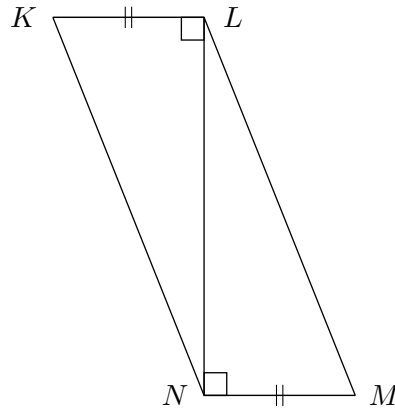
- (d) A gold ingot in the form of a rectangular prism with dimensions 8 cm, 6 cm and 3 cm is melted down. How many cylindrical coins of diameter 1.5 cm and thickness 2 mm can be made from this gold?

3**Question 5** (10 Marks)

Commence a NEW page.

Marks

- (a) In
- $\triangle KLN$
- and
- $\triangle LNM$
- ,
- $KL = NM$
- and
- $\angle KLN = \angle LNM = 90^\circ$
- .

4Prove that $\triangle KLN \equiv \triangle LNM$.

- (b) Simplify fully:

i. $\frac{x^2 + 3x + 2}{x + 2}$.

2

ii. $\frac{2}{x + 1} \times \frac{5x + 5}{4}$.

2

iii. $\frac{2}{x^2y} - \frac{3}{xy^2}$.

2**End of paper.**

$$\begin{aligned} \text{Q1) a) i) } & (5y-2)(2y+3) \\ & = 10y^2 + 15y - 4y - 6 \\ & = 10y^2 + 11y - 6 \end{aligned}$$

$$\begin{aligned} \text{ii) } & (3b+1)(3b+1) \\ & = 9b^2 + 6b + 1 \end{aligned}$$

$$\text{b) i) } y(3y-2)$$

$$\text{ii) } (a+b)(x+y)$$

$$\text{iii) } (d-a)(d+a)$$

$$\text{iv) } (a+5)(a-3)$$

$$\begin{aligned} \text{v) } & (x^2+1)(x^2-1) \\ & = (x^2+1)(x+1)(x-1) \end{aligned}$$

$$\begin{aligned} \text{c) i) } & \frac{21x + 10x}{14} \\ & = \frac{31x}{14} \end{aligned}$$

$$\begin{aligned} \text{ii) } & \frac{\cancel{3}(x+7)}{\cancel{3}} \\ & = x+7 \end{aligned}$$

$$\begin{aligned} \text{iii) } & \frac{1(5b+4b)}{2(5a+4b)} \\ & = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{Q2) a) i) } & A = 4 \left(\frac{7+15}{2} \right) \\ & = 44 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{ii) } & A = \frac{1}{2} 10.18 \\ & = 90 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{b) } & A = \pi r^2 \\ 3.0512 & = \pi r^2 \\ r^2 & = \frac{3.0512}{\pi} \\ & = 0.9712 \dots \\ r & = 0.985 \\ & = 0.99 \text{ to } 2 \text{ d} \end{aligned}$$

$$\begin{aligned} C & = 2\pi r \\ & = 2\pi \times 0.99 \\ & = 6.19 \text{ m}^2 \\ & \quad 6.22 \end{aligned}$$

$$\begin{aligned} \text{c) } & A = \frac{210}{360} \pi r^2 \\ & = \frac{7}{12} \times \pi \times 9^3 \\ & = \frac{21\pi}{4} \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{d) } & A = \pi r_2^2 - \pi r_1^2 \\ & = \pi 25 - \pi 4 \\ & = 21\pi \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{e) } & V = 8(\pi (10.4)^2 - \pi (3.7)^2) \\ & = 2374.29 \text{ cm}^3 \end{aligned}$$

Q3) a) i) Germany

ii) 213

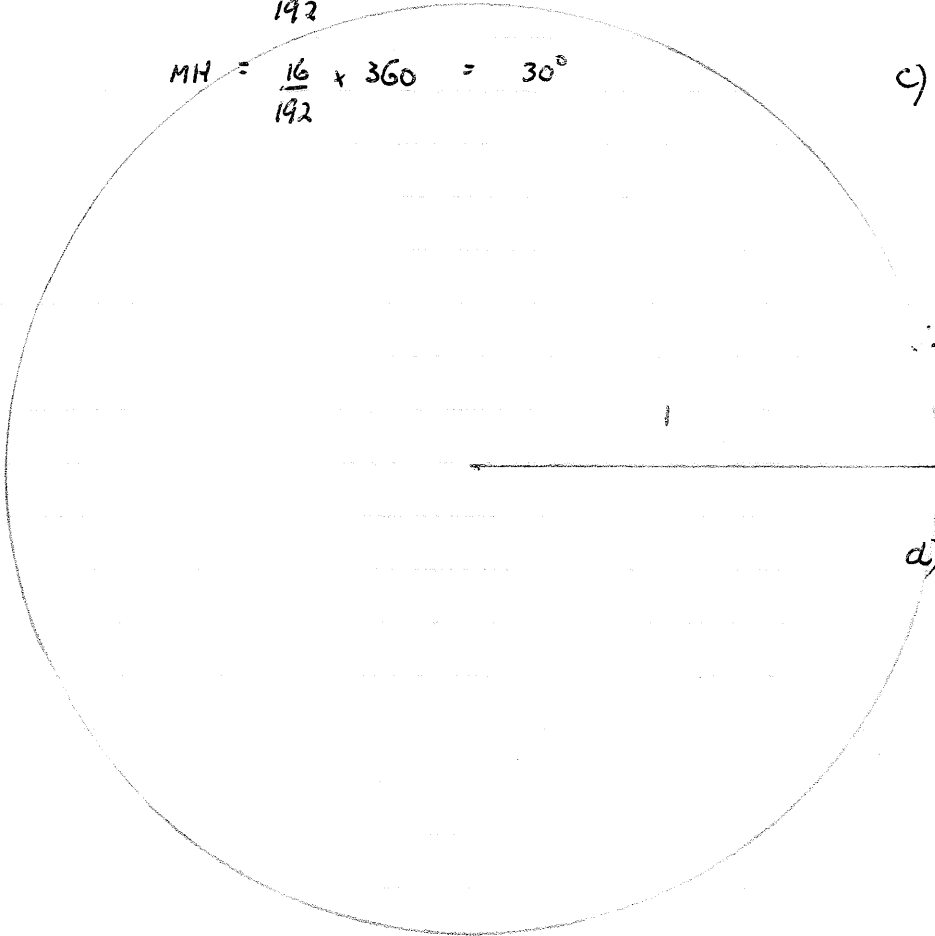
iii) 78 p/km²

$$b) N = \frac{32}{192} \times 360 = 60^\circ$$

$$M = \frac{80}{192} \times 360 = 150^\circ$$

$$B = \frac{64}{192} \times 360 = 120^\circ$$

$$MH = \frac{16}{192} \times 360 = 30^\circ$$



c) i) 50° F

ii) 23° F

iii) 0° C

iv) -17.5° C

$$Q4) a) V = \pi r^2 h$$

$$= \pi 25 \left(\frac{7+13}{2} \right)$$

$$= 250 \pi \text{ cm}^3$$

$$b) i) V = \pi r^2 h$$

$$= \pi 64 \cdot 10$$

$$= 640 \pi \text{ cm}^3$$

~~$$ii) S.A. = 2\pi r^2 + 2\pi r h$$~~
~~$$= 2\pi 64 + 2\pi \cdot 8 \cdot 10$$~~
~~$$= 288 \pi \text{ cm}^2$$~~

$$c) V = \pi r^2 h$$

$$300 = \pi r^2 \cdot 2$$

$$25 = \pi r^2$$

$$r = 5$$

$$\therefore S.A. = 2\pi r^2 + 4\pi r h$$

$$= 50 + 240$$

$$= 290 \text{ cm}^2$$

$$d) V = 8 \times 6 \times 3$$

$$= 144 \text{ cm}^3$$

$$V_{\text{coin}} = \pi r^2 h$$

$$= \pi \left(\frac{3}{4} \right)^2 \times 0.2$$

$$= .35343$$

$$\text{Number} = \frac{V}{V_{\text{coin}}}$$

$$= 407.4$$

Q5)

a) In Δ 's KLN & KNM

$$\angle KLN \equiv \angle LNM \text{ (given)}$$

$$\angle L = \angle M \text{ (given)}$$

 LN is common

$$\therefore \Delta KLN \equiv \Delta LNM \text{ (S.A.S.)}$$

$$b) \quad i/ \quad \frac{(x+2)(x+1)}{x+2}$$

$$= x+1$$

$$ii/ \quad \frac{2}{x+1} \times \frac{5(x+1)}{4}$$

$$= \frac{5}{2}$$

$$iii/ \quad \frac{2}{x^2y} - \frac{3}{xy^2}$$

$$= \frac{2y - 3x}{x^2y^2}$$